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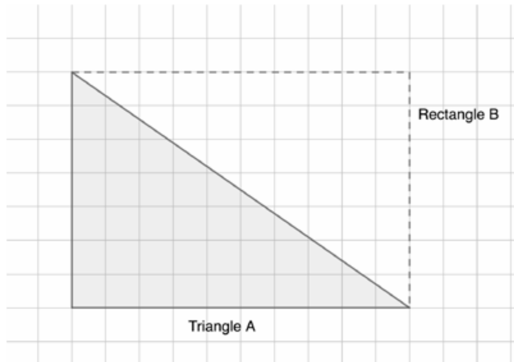
Lesson 1 Homework Period___________________

Figure 1 shows **right triangle** A, within Rectangle B.



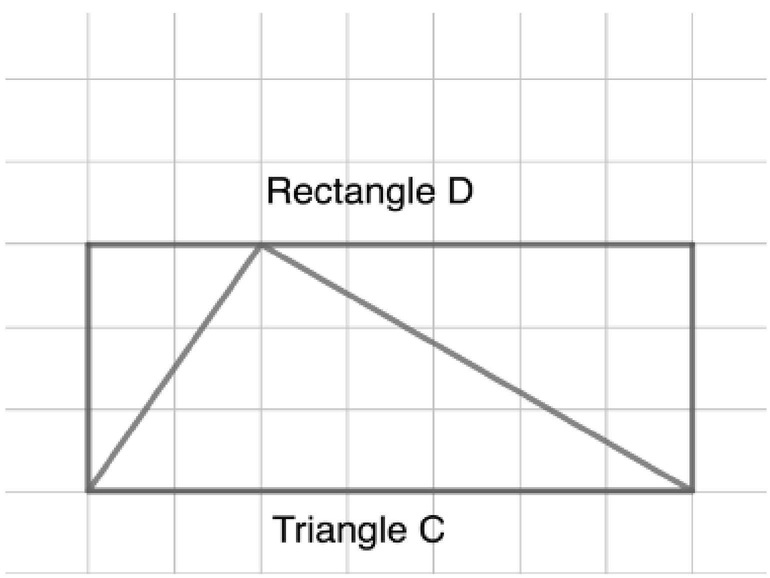
Use the figure to answer questions 1-3. Each square of the grid is 1 cm by 1 cm.

1. It takes _____ triangles to form Rectangle B.

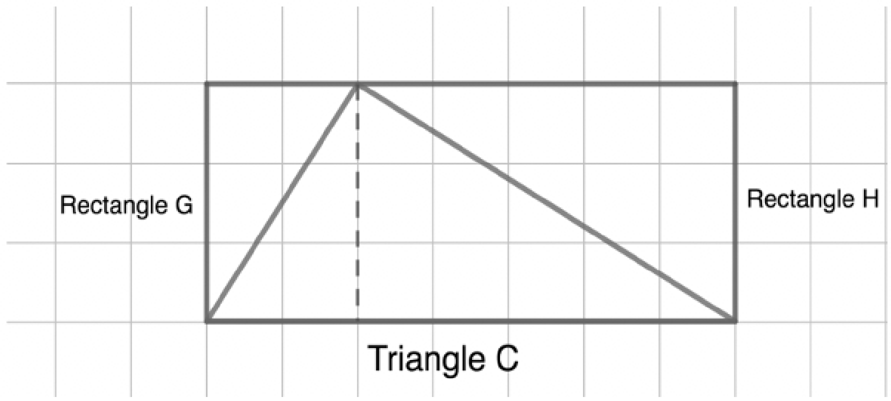
2. The area of Rectangle B is _______ cm^2.

3. The area of Triangle A is _______ cm^2.

The figure shows an **acute triangle**, Triangle C. Rectangle D is drawn around Triangle C.



The figure can be broken into smaller rectangles, Rectangle G and Rectangle H, and two right triangles.



Use the figures to answer questions 4-7. Each square of the grid is 1 cm by 1 cm.

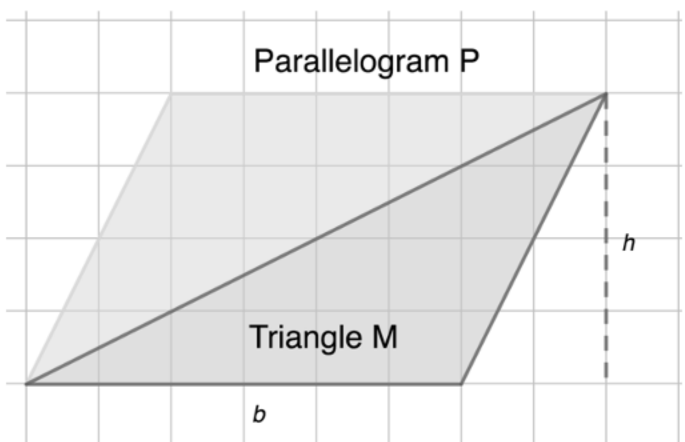
4. The area of Rectangle G is _______ cm^2.

5. The area of Rectangle H is _______ cm^2.

6. The area of a right triangle is ______ of the area of a rectangle.

7. The total area of Triangle C is _______ cm^2.

Triangle M is an **obtuse triangle**. Parallelogram P is formed by duplicating Triangle M and rearranging it.



Use the figures to answer questions 8-10. Each square of the grid is 1 cm by 1 cm.

8. The area of Parallelogram P is _______ cm^2.

9. Triangle M represents ______ of the figure of Parallelogram P.

10. The area of Triangle M is _______ cm^2.

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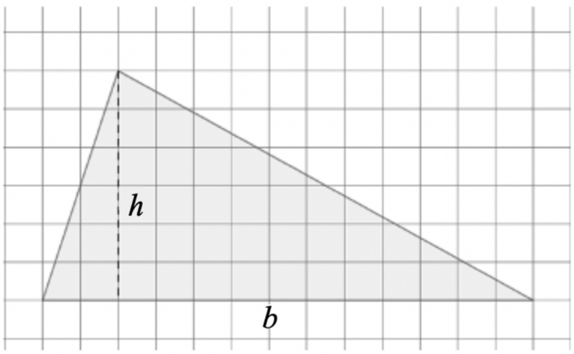
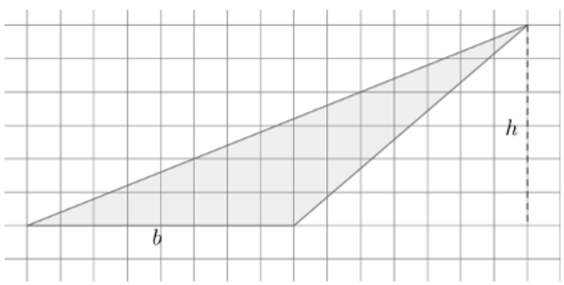
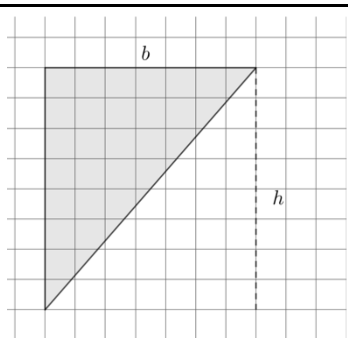
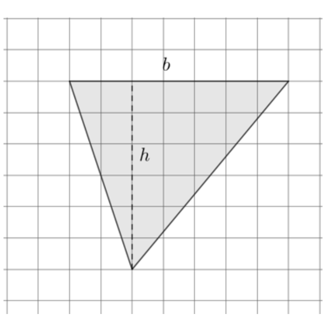
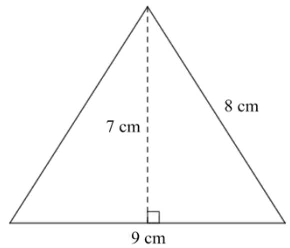
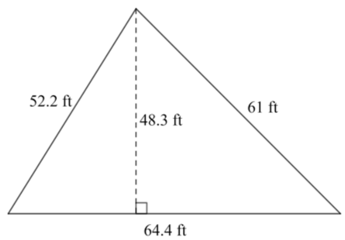
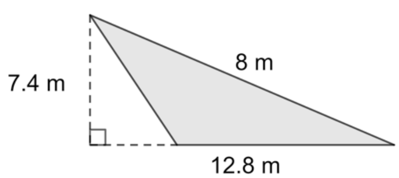
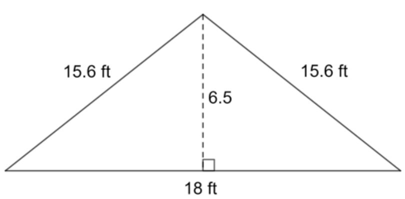
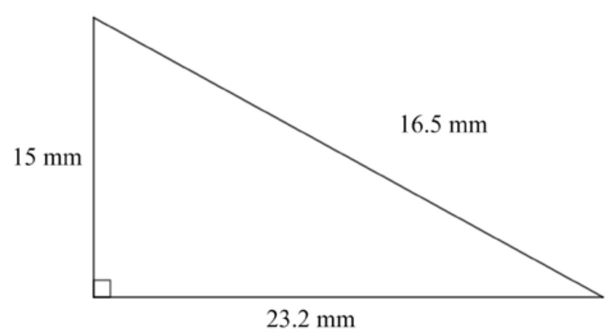
Four triangles are shown on the grids. For each triangle, a base and its corresponding height are labeled. The side length of each square is 1 centimeter (cm). Find the area of each triangle. Show your work and include units.

1.		Area = _____________
2.		Area = _____________
3.		Area = _____________
4.		Area = _____________

For each triangle below, circle the base measurement and put a square around the height measurement. Then, use the base and height to find the area of each triangle. Show your work and include units.

5.		Area = _____________
6.		Area = _____________
7.		Area = _____________
8.		Area = _____________

Janelle found the area of the triangle shown.
Her work is also shown.
Area = $23.2 \times 16.5 \times 0.5$
Area = 191.4 mm^2



9. Explain why Janelle's answer is incorrect.

10. Find the correct area of the triangle.

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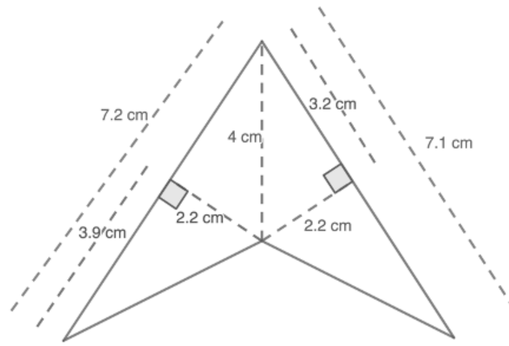
A quadrilateral with measurements in centimeters (cm) is shown for questions 1–3.

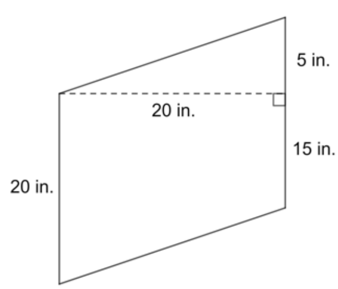
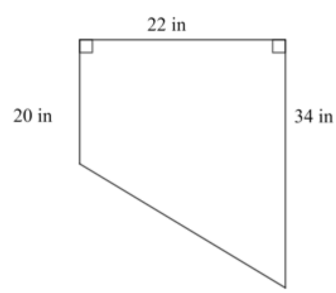
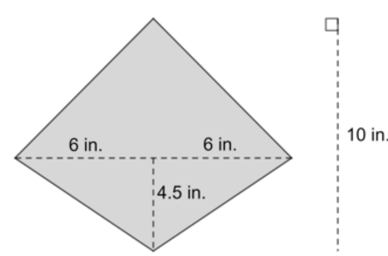


1. Decompose the quadrilateral into triangles. Sketch the decomposed shapes here.

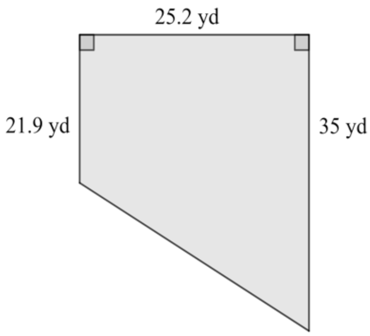
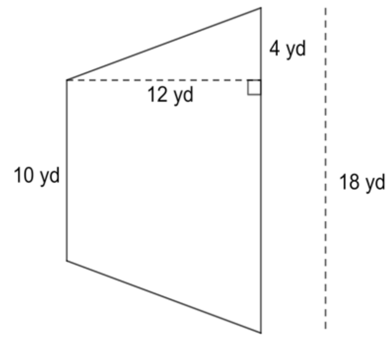
2. Using the measurements given in the original image, label the base and height of each triangle.

3. Find the area of the original quadrilateral by finding the sum of the areas of the decomposed triangles. Include units.

Decompose each quadrilateral to find the area. Lengths are shown in inches (in.).

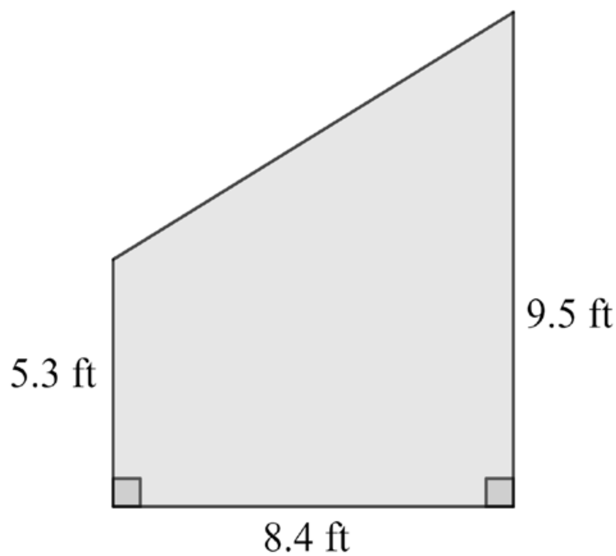
4. 	5. 	6. 
Area = ___________	Area = ___________	Area = ___________

Decompose each quadrilateral to find the area. Lengths are shown in yards (yd).

7. 	8. 
Area = ___________	Area = ___________

Sal made a sketch of a wall he needs to paint. The sketch of the wall is shown with measurements in feet for questions 9–10.

9. The home improvement store is offering a sale on their quarts of paint. A quart of paint covers 100 square feet of wall. Will Sal have enough paint if he buys one quart?



10. Determine the amount of paint, in square feet, Sal is short or has left over.

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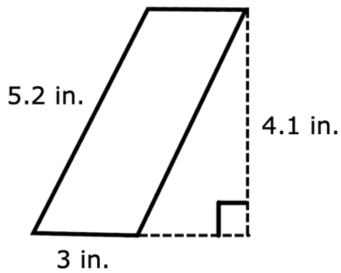
A parallelogram with dimensions labeled is shown below for questions 1-2.



1. Which dimension is not needed to find the area of the parallelogram? Explain.

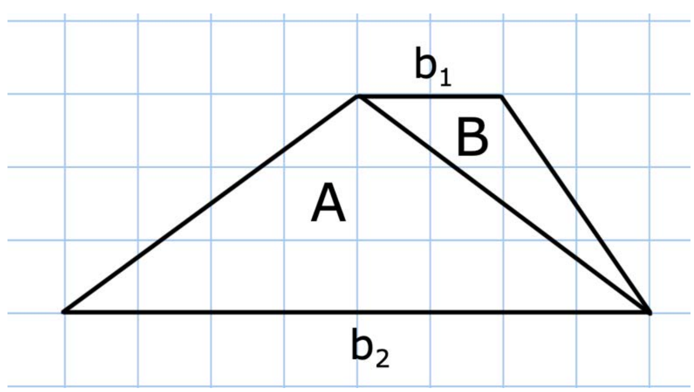
2. Fill in the blanks to find the area of the parallelogram.

   Area of parallelogram = $b \cdot h$

   Area = ____ ·____

   Area = ____ in.^{2}

A trapezoid can be decomposed into two triangles with the same height. Use the diagram below for questions 3-6.



3. The base of triangle A, b_2, is _____ units and its corresponding height is _______ units. Therefore, the area or triangle A is ________.

4. The base of triangle B, b_1, is _____ units and its corresponding height is _______ units. Therefore, the area or triangle B is ________.

5. The area of the trapezoid is ___________ units^2.

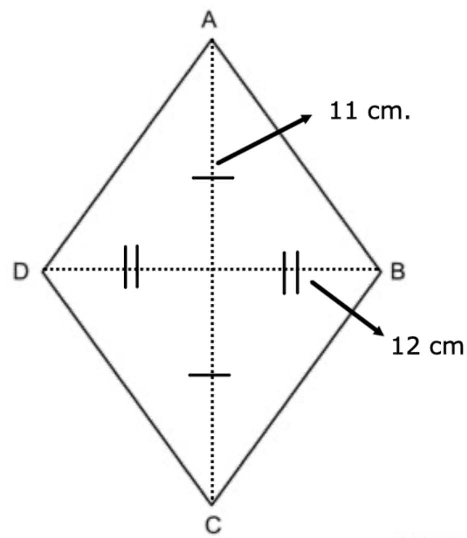
6. Use the formula for area of a trapezoid to verify your solution is correct.

Area = $\frac{1}{2}h(b_1 + b_2)$.

Area = $\frac{1}{2}$ ___ (___+ ___)

Area = ______ units^2

Rhombus $ABCD$ is shown. Diagonal BD is 12 centimeters and diagonal AC is 11 centimeters. Use the diagram below for questions 7-10.



7. The base of triangle ABC is _____ cm. and its corresponding height is _______ units. Therefore, the area of triangle ADC is ________.

8. The base of triangle ADC is _____ cm. and its corresponding height is _______ units. Therefore, the area or triangle ADC is ________.

9. The area of the rhombus is ___________ cm^2.

10. Use the formula for area of a rhombus to verify your solution is correct.

Area = $\frac{1}{2}(d_1)(d_2)$.

Area = $\frac{1}{2}$(___)(___)

Area = ______ units^2

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1. The formula to find the area of a parallelogram is

2. The formula to find the area of a trapezoid is

3. The formula to find the area of a rhombus is

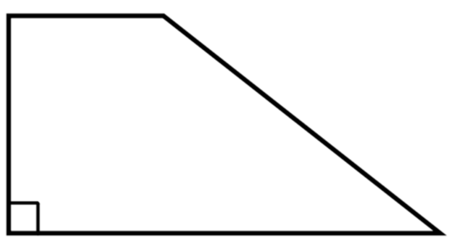
For questions 4-6, label each figure with the given dimensions, then calculate the area using the appropriate formula.

4. $b = 12.3$ in
$h = 7.1$ in

Area =

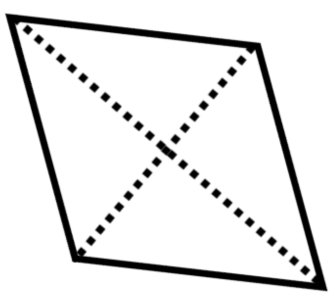
5. $b_1 = 3\frac{1}{2}$ in
$b_2 = \frac{7}{4}$ in
$h = 2$ in

Area =



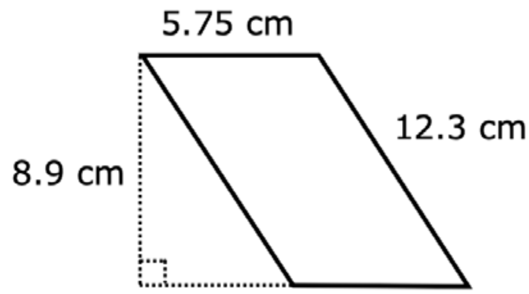
6. $d_1 = 7$ in
$d_2 = 5$ in

Area =

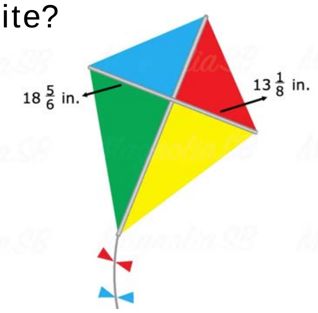


7. Zoey made an error in her work. Identify the error and then find then find the area of the parallelogram.

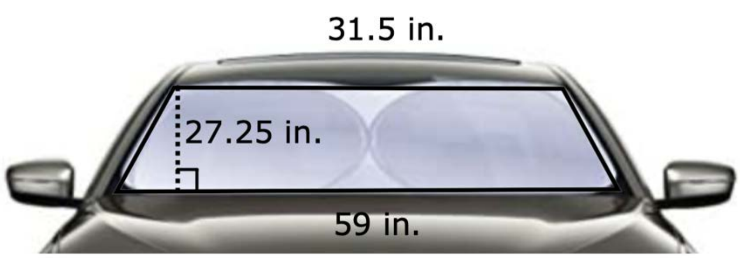
Area = $(5.75)(12.3)$
Area = $70.725\ \text{cm}^2$



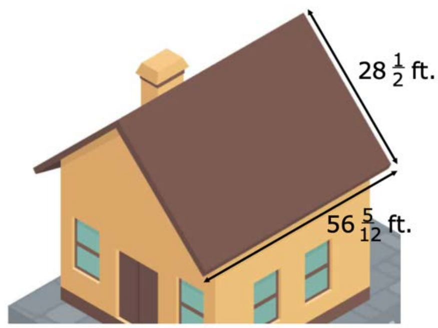
8. Kai and his father are constructing a kite to fly at the beach that is in the shape of a rhombus. The spars on the kite measure $13\frac{1}{8}$ inches and $18\frac{5}{6}$ inches. How many square inches (in^2) of fabric will be needed to construct the kite?



9. The windshield of a car is in the shape of a trapezoid. Find the area of the glass used to make the windshield.



10. Marc needs a new roof on his house and needs to measure how many roofing tiles are needed to get an accurate estimate from the roofing company. The roof has two sides. Each side is in the shape of a parallelogram with a base of $56\frac{5}{12}$ feet and a height of $28\frac{1}{2}$ feet. Find the how many square feet of roofing material is needed.



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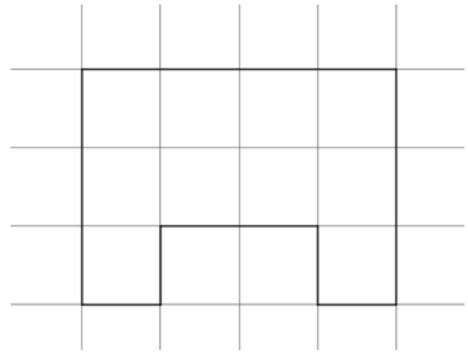
A composite figure is shown for questions 1–3. The length of each square is 1 inch (in.).

1. Draw lines to decompose the figure into rectangles.

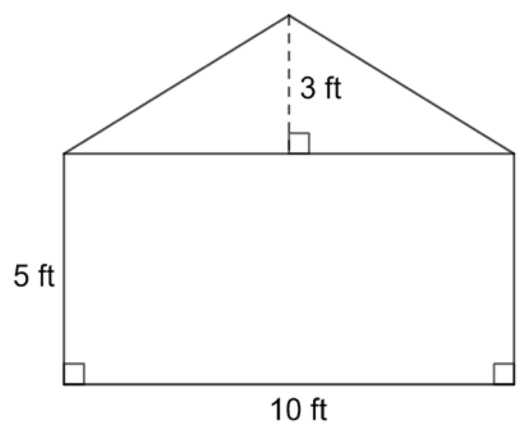


2. Find the area of each rectangle. Include units.

3. Find the sum of the areas of the rectangles to find the area of the composite figure. Include units.

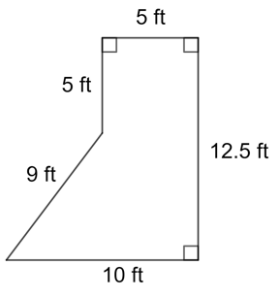
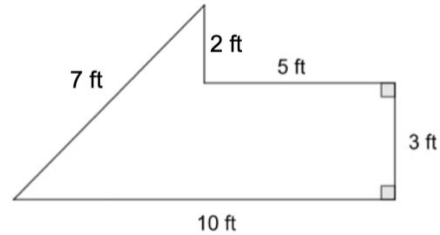
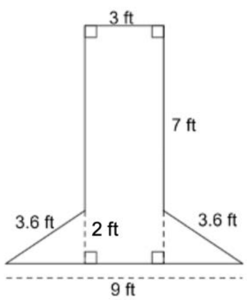
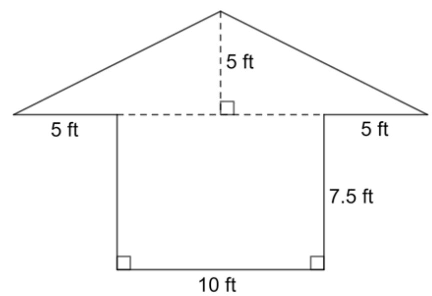
A composite figure measured in feet (ft) is shown for questions 4–5.

4. Find the area of the triangle and the rectangle. Include units.

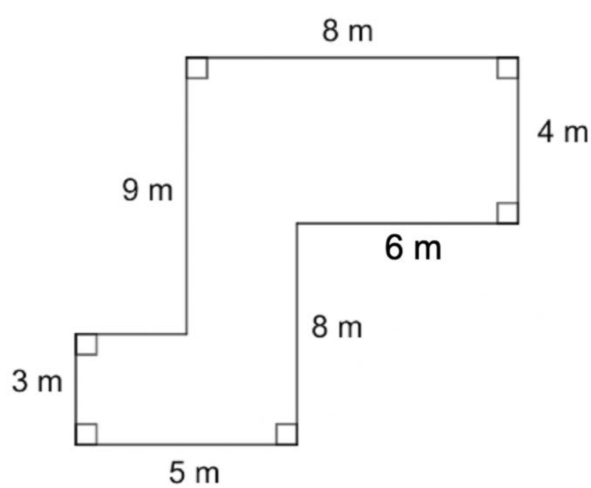


5. Find the sum of the areas of the rectangle and triangle to find the area of the composite figure. Include units.

For each composite figure, find the area in square feet.

6. 	7. 
Area = _________	Area = _________
8. 	9. 
Area = _________	Area = _________

10. The local university is planting a garden. A drawing of the garden is shown. What is the area of the garden in square meters?



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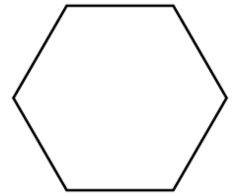
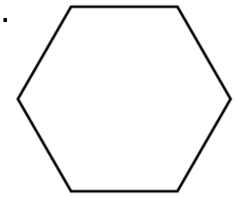
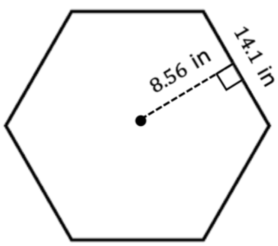
1. Decompose the regular hexagon into only triangles.



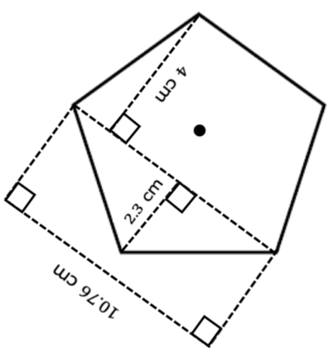
2. Decompose the regular hexagon into triangles and quadrilaterals.



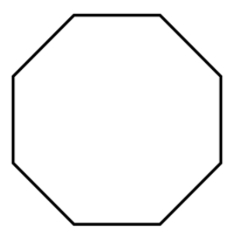
3. A hexagon is given in inches (in). Each side of the hexagon has a length of 14.1 inches and a perpendicular distance from the center to the midpoint of a side to be 8.56 inches. Find the area of the regular hexagon rounded to the nearest square inch.



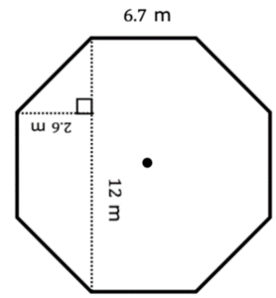
4. Find the area of the regular pentagon that has a side length of $5\frac{1}{2}$ cm.



5. Decompose the regular octagon into three quadrilaterals.

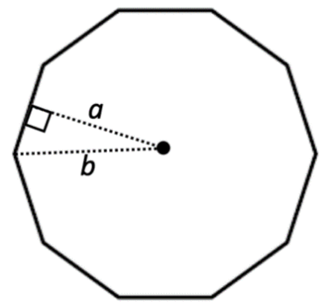


6. An octagon is shown with lengths in meters (m). What is the area of the octagon?



7. Label the regular decagon with the following dimensions.
Side length = 6.2 in.
$a = 7.4$ in.
$b = 9.22$ in.

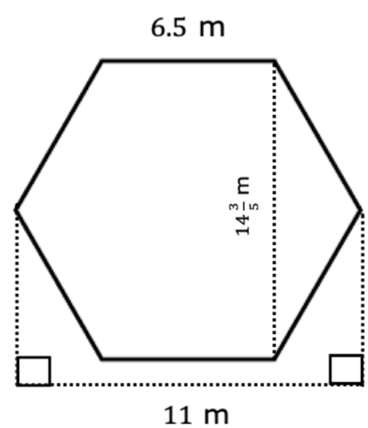
Which of the dimensions is not needed to find the area of the decagon?



8. Find the area of the decagon from Question 7 in square inches.

9. Myles found the area of the hexagon. His work is shown.

Myles' Work
Area of Trapezoid = $7.3(11 + 6.5)$ $A = 127.75$ m^2 Area of hexagon = $2(127.75)$ $A = 255.5$ m^2



Write a note to Myles explaining the error he made.

10. Find the area of the hexagon in square meters.

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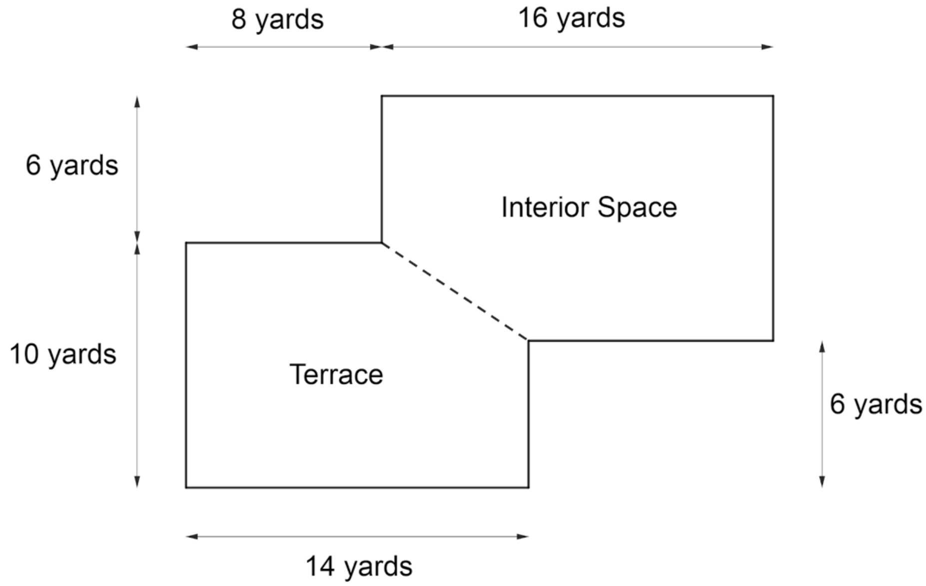
Lesson 8 Homework Period__________________

A local restaurant consists of both interior space and an outdoor terrace.

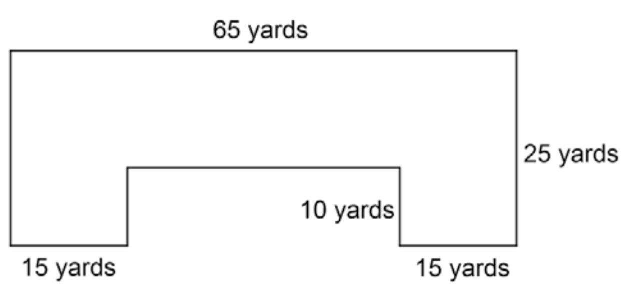


1. Decompose the floorplan into triangles and quadrilaterals.

2. Find the area of the terrace.

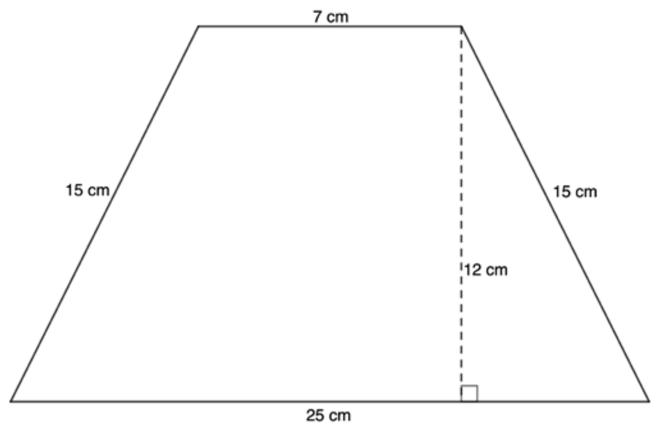
3. Find the area of the interior space.

4. Find the area of the entire restaurant.



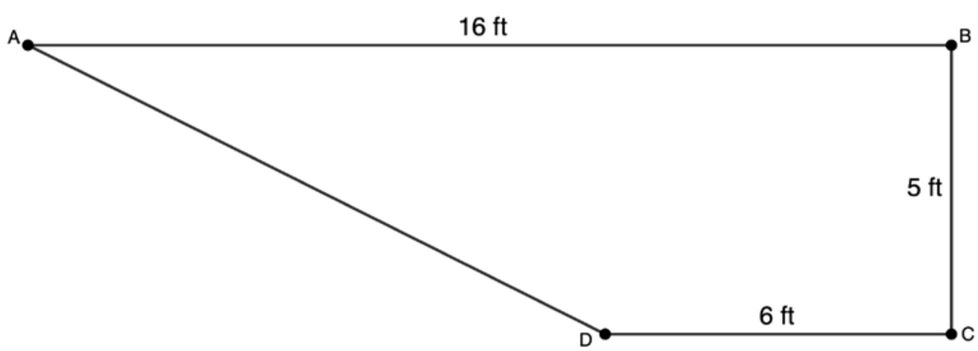
5. Decompose the image into quadrilaterals.

6. Find the area of the composite figure.



7. Decompose the image into triangles and quadrilaterals.

8. Find the area of the image.



9. Decompose the image into triangles and quadrilaterals.

10. Find the area of the composite figure.